

Algebra 1
Unit 9
Lesson 1 Practice

Name Answer key
Date _____
Period _____

Use the equation below to answer questions 1 through 3.

$$y = a(x - 1)(x + 3)$$

1. Write the x -intercepts of the quadratic equation.

$$\begin{array}{lll} x - 1 = 0 & x + 3 = 0 & (1,0) \\ x = 1 & x = -3 & (-3,0) \end{array}$$

2. Solve for a , given that $(-2, -6)$ is a point on the quadratic's graph.

$$\begin{array}{ll} -6 = a(-2 - 1)(-2 + 3) & a = 2 \\ -6 = -3a & \end{array}$$

3. Using the value of a found in question 2, write the quadratic equation in factored form.

$$y = 2(x - 1)(x + 3)$$

4. Rewrite the quadratic equation in standard form.

$$\begin{array}{l} y = 2(x^2 + 2x - 3) \\ y = 2x^2 + 4x - 6 \end{array}$$

The x -intercepts and a point on the parabola are given for several quadratic functions. For each function, show how to find the value of a and write each quadratic in factored form.

	x -intercepts	Point	Steps to Find " a "	Quadratic Equation in Factored Form
5.	$(x - 1)(x - 4)$ $(1,0)$ and $(4,0)$	$(-2, 6)$	$6 = a(-2 - 1)(-2 - 4)$ $6 = a(-3)(-6)$ $6 = 18a$ $\frac{1}{3} = a$	$y = \frac{1}{3}(x - 1)(x - 4)$

	x-intercepts	Point	Steps to Find "a"	Quadratic Equation in Factored Form
6.	$(x + 5)(x + 2)$ $(-5, 0)$ and $(-2, 0)$	$(4, -3)$	$-3 = a(4 + 5)(4 + 2)$ $-3 = a(9)(6)$ $-3 = 54a$ $\frac{-1}{18} = a$	$y = -\frac{1}{18}(x + 5)(x + 2)$
7.	$\left(x - \frac{1}{2}\right)(x - 3)$ $\left(\frac{1}{2}, 0\right)$ and $(3, 0)$	$\left(-1, \frac{3}{2}\right)$	$\frac{3}{2} = a\left(-1 - \frac{1}{2}\right)\left(-1 - 3\right)$ $\frac{3}{2} = a(-1.5)(-4)$ $\frac{3}{2} = 6a$ $\frac{1}{4} = a$	$y = \frac{1}{4}(2x - 1)(x - 3)$
8.	$(x + 0)(x - 5)$ $(0, 0)$ and $(5, 0)$	$(1, -24)$	$-24 = a(1)(1 - 5)$ $-24 = a(-4)$ $-24 = -4a$ $6 = a$	$y = 6x(x - 5)$
9.	$(x + 3)(x + 3)$ $(-3, 0)$ and $(-3, 0)$	$(-2, 2)$	$2 = a(-2 + 3)^2$ $2 = a(1)^2$ $2 = a$	$y = 2(x + 3)^2$
10.	$(x - 2)(x + 2)$ $(2, 0)$ and $(-2, 0)$	$(4, 3)$	$3 = a(4 - 2)(4 + 2)$ $3 = a(2)(6)$ $3 = 12a$ $\frac{1}{4} = a$	$y = \frac{1}{4}(x - 2)(x + 2)$